Thomas GARABEDIAN

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Education

Institut Polytechnique de Paris, École Polytechnique — Sep. 2026
Master 2 Data Science (M2DS) — *Palaiseau, France*

- Targeted coursework: Machine Learning, Deep Learning, GNNs, LLMs, Computer Vision.

Institut Polytechnique de Paris, Télécom SudParis — 2024 – 2027
Engineering Degree in Digital Technologies — *Évry, France*

- **GPA: 3.7/4.0**, top 15% among 220 students.
- Relevant coursework: Applied Statistics, Optimization, Signal Processing, Data Mining, Cryptography, Secure Communications, Data Analysis, Stochastic Processes.

CPGE Lycée Thiers & Prytanée National Militaire — 2021 – 2024
Intensive undergraduate program preparing for French engineering school, PCSI then PC track — *France*

Experience

CIEDS IP Paris Project | Drone Swarm Automation — Jan. – June 2026
ENSTA Paris — *Palaiseau, France*

- Designed a computer vision pipeline to detect vehicles and buildings in aerial imagery, in a multi-drone surveillance context. Presented the results to CIEDS IP Paris.
- Contributed to the swarm coordination logic: target assignment, trajectory planning and scenario simulation. Compared object detection models: CNN, YOLO, DETR.

Engineering Internship | ALSEAMAR, Alcen Group — June – Aug. 2025
Embedded systems, RF and industrial software — *Six-Fours, France*

- Developed control software for a high-frequency power amplifier used in an industrial environment and for submarine antenna testing.
- Implemented equipment communication, supervision, acquisition, data logging and diagnostic modules.
- Carried out validation tests to improve system control reliability and measurement traceability.
- Worked at the interface of Python, embedded systems, RF electronics, signal processing and industrial constraints.

Special Military Preparation — 2023
École du Génie — *Thorée-les-Pins, France*

- Intensive training in leadership, decision-making under constraints and defense fundamentals.

AI & Data Science Projects

HAKS Hackathon, Airbus x IBM | 1st Prize, €2,500 — June 2026
Télécom Paris — *Palaiseau, France*

- Developed a corrosion prediction model for an aeronautical use case, combining FT-Transformer and XGBoost.
- Built an end-to-end machine learning pipeline, from data preprocessing and feature engineering to model comparison, calibration and reliability-focused evaluation.
- Optimized and probabilistically evaluated the model with a Brier Score of 0.150.

D4GEN Hackathon, Genopole x AWS | 1st Prize, €10,000 — May – June 2026
Genopole d'Évry — *Évry, France*

- Designed a human-in-the-loop tool to accelerate expert annotation for animal species monitoring.
- Built a pipeline combining unsupervised clustering, clip export, spectrograms and an annotation API.
- Fine-tuned SAM 3 for segmentation and visual identification support in a weakly labeled data context.

Skills

- **AI & Computer Vision:** object detection, segmentation, classification, CNN, YOLO, DETR, SAM 3, Transformers, FT-Transformer, XGBoost, LightGBM, CatBoost.
- **Model Evaluation:** IoU, precision, recall, error analysis, probabilistic calibration, Brier Score, WAPE.
- **Languages:** French (Native), English (C1), Spanish (C1).
- **Sports:** Competitive tennis, official referee, ranked first in training cohort.